

Sound Biosynthetic Design over Iterative Grammars: A Realizability Geometry for Fungal Polyketides

Brage Eilertsen
bragee2002@gmail.com
Affiliation to be completed

May 2026 — working draft (design reframe); §6 first per the composition plan

Abstract

[Stub — written last. Spine: wrong object (BGC $\rightarrow$ structure regression, the 51–68% ceiling) $\rightarrow$ program space $y = \text{Exec}_\Theta(z; G)$ run forward / inverse / backward $\rightarrow$ a sound triage tool with named limits. Most of the abstract is the framing-note one-paragraph thesis.]

1 Introduction

[Stub — drafted after §6 and §3. Wrong frame $\rightarrow$ right frame (program space; inference and design); the one-sentence thesis; architecture figure. ¶2 lever: Cox (2023) identifies machine learning as the route forward where rational HR-PKS engineering “will remain extremely difficult” — taken as a design constraint, not a validation. ¶3: structure elucidation here proceeds by expression + NMR with MS as triage; we build a triage tool grounded in metabologenomics observables and name explicitly where triage is insufficient.]

2 Substrate: a sound executor and verifier

[Stub — edited from the inverse-compiler manuscript (`paper/main.tex`). Sound deterministic executor + beam verifier; the three-state verdict (VERIFIED / UNDER-OBSERVED / OUT-OF-GRAMMAR); guarantees relative to operator library, grammar, observables, mass tolerance, adduct model.]

3 Characterizing the natural manifold

[Stub — drafted second, inheriting §6’s phrasing. Opener (does the reframe and inoculates against being read as a limitations chapter): “the natural fungal PKS manifold has 8 exactly-reachable cores, a 0.57 control-axis ceiling, and a tailoring-permissive interior with 64% conditional reconstructibility — this section characterizes its shape.” $8/326 =$ the manifold’s boundary condition (not a prediction ceiling); the structural-similarity proxy critique as honest defensive work ($8 \rightarrow 160$); the 0.57 wall as the axis along which the interesting designs live (not where prediction fails). Refers forward to §6.]

4 Generality across grammars

[Stub — ONE PAGE. PKS, NRPS, and a typed PKS–NRPS hybrid behind one interface; the load-bearing point is that the observability and verdict layers are family-agnostic, not the demo

counts. Reviewers wanting more cross-grammar work are pointed at the repository.]

5 Mining: genome and metabolome

[Stub — ONE PAGE. The $BGC \leftrightarrow \text{peak}$ assignment graph; structure verdict $\perp$ attribution verdict; one real blind cluster $\rightarrow$ mass case (6-MSA, MIBiG BGC0001275). A bridge, not a pillar.]

6 Sound biosynthetic design over the iteration grammar

Running the executor forward turns a program into a structure; running the verifier inverse turns a structure into the set Z^* of programs that could have produced it. Here we run the grammar *backward*: given a desired, possibly non-natural target, we place it on a *realizability geometry* relative to the natural manifold, and we rank the experiments that would confirm it. Two results make this more than a restatement of the diagnostic half. First, the design space decomposes onto two axes that map to fundamentally different wet-lab capabilities. Second—and this is what keeps the contribution honest—the cost on those axes is the field’s *measured* difficulty of fungal PKS engineering, curated from primary literature, not an assumption we supply.

6.1 The realizability geometry: two axes

A design target is a program z_d ; its product is $y_d = \text{Exec}_\Theta(z_d; G)$. We score z_d by its edit distance to the nearest natural cluster on the manifold, decomposed into two typed axes:

- **Structural** edits change the *domain content*: add a reductive domain (`add_reductive`), swap the starter or extender (`starter`, `extender`), reprogram the release cyclase (`release`), or insert a methyltransferase de novo (`add_cmt`). These map onto bacterial modular-PKS engineering.
- **Control** edits change the *iteration program* with the domain set fixed: which cycle a present domain fires on (`reduction`, `c_methyl`) or the iteration count (`cycle_add`, `cycle_remove`). This is the iteration-grammar frontier—precisely the 100% $\rightarrow$ 57% wall of the cross-taxa transfer probe (§3), now read as a design dimension rather than a prediction limit.

A design at coordinate (2, 1) (two structural, one control edit) is a different proposition from one at (3, 0): the first requires reprogramming the iteration itself, the second is documented domain engineering. Each target’s most-realizable route is the minimum-cost edit path to a natural cluster, and its *verdict*—NATURAL, ENGINEERABLE, FRONTIER, or SPECULATIVE—is read off that path. Crucially, the control axis *only exists as a design dimension* for a machine that distinguishes per-cycle firing within a fixed domain set; a tool without the iteration grammar cannot represent the edit “reprogram the ketoreductase at cycle 3,” let alone cost it (§6.6).

6.2 A literature-grounded, super-additive cost

The design enumeration bounds the cost model to the nine edit kinds that actually appear in the 1-edit neighbourhood, and we curated each from primary sources rather than scraping abstracts (two tier rulings turned on facts a search snippet stated incorrectly). The structural edits inherit the mature combinatorial-biosynthesis canon: acyltransferase/extender swaps and loading-module/starter swaps are documented (AT-engineering review, 2018; Erythromycin analogues, 1997), as are reductive-loop transplants (Reductive-loop swaps, 2008); de novo C-methyltransferase insertion has no

[Figure: the 1-edit design neighbourhood of 6-MSA, as a ranked build table. Source: `scripts/worked_design_6msa.py`. Columns: (S,C) edit decomposition; verdict; predicted monoisotopic mass; verifying observable. Two labelled blocks (engineerable / frontier), executor-verifiable designs leading.]

Figure 1: The worked 6-MSA design neighbourhood: design distance and verification plan on one line.

precedent (the methylation literature reprograms a *present* C-MeT, never inserts one) and is scored SPECULATIVE.

The control edits are the unsolved frontier. Cox’s review of iterative highly-reducing PKS programming (Cox, 2023) states plainly that rational efforts “will remain extremely difficult” because the programme is “an emergent property... highly unpredictable.” Yet a *single* edit beside a closely related cluster is achievable: domain swaps between the tenellin and desmethylbassianin synthases mapped methylation-pattern and chain-length changes cleanly (Fisch et al., 2011). We encode exactly this shape. Structural edits sum linearly by tier; control edits are **super-additive**—quadratic in the number k of control edits in a design:

$$\text{cost}(z_d) = \underbrace{\sum_{e \in \text{struct}} \tau(e)}_{\text{linear}} + \underbrace{\left(\sum_{e \in \text{ctrl}} \tau(e) \right) \cdot k_{\text{ctrl}}}_{\sim k^2, \text{ super-additive}} \quad (1)$$

where τ is the curated per-edit tier. Equation (1) reduces to the linear tier cost at $k = 1$ (the Fisch–Cox single-edit regime) and grows quadratically as edits stack (the Cox emergent regime). The cost model is therefore not asserted: it is the published difficulty landscape of fungal PKS engineering, written into the objective. This is the difference between “we built a design engine” and “we built a design engine whose cost model is the measured difficulty of the field.”

6.3 The worked neighbourhood of 6-MSA

Figure 1 walks one clean template—the 6-methylsalicylic acid synthase, the cluster with a verified blind real-mass match (§5)—end to end. Each row of its 1-edit neighbourhood is a build proposal carrying its (structural, control) coordinate, realizability verdict, predicted monoisotopic mass, and the observable that would confirm it. The neighbourhood separates cleanly into an **engineerable** block (single structural edits—starter swaps such as acetyl→hexanoyl at mass 244.13, release reprograms—all confirmable by accurate mass) and a **frontier** block (single control edits such as reprogramming the cycle-3 reduction), with coverage-limited out-of-grammar designs sunk below the headline. One row reads, “one documented edit (acetyl→hexanoyl) from the 6-MSA synthase, mass 244.13, verifiable by accurate mass”; another, “one iteration-program edit (reprogram cycle-3 reduction), mass 172.07, needs MS/MS.” Design distance, the predicted observable, and a verdict on whether triage suffices or structure elucidation is required—on the same line.

6.4 Control-axis dominance

Across the 1-edit neighbourhood of the natural manifold the control axis dominates: **69** designs lie on the frontier (involve a control edit) against **49** engineerable (structural-only), with 24 speculative. That ratio is itself a product of the literature curation—the raw, uncured count was a starker 93:34, and grounding the edit costs (demoting release to a structural domain swap, raising de novo C-MeT and iteration-count edits) softened it to a defensible margin. Fungal biosynthetic *design* is mostly iteration-program editing: the axis only a sound per-cycle executor can express.

[Figure: two decision trees side by side. Source: `scripts/planner_tree_figure.py` $\rightarrow$ `results/planner_tree_{6msa,bab}.mmd`. Left: 6-MSA, root decision (*run mass?*) $\rightarrow$ confirmed $\Rightarrow$ verified-by-triage; refuted $\Rightarrow$ out-of-grammar. Right: BAB, mass $\rightarrow$ MS/MS exhausted $\Rightarrow$ elucidation-required; refuted leaves drawn first-class at each level.]

Figure 2: The triage planner doing its job (6-MSA) and naming its own limit (BAB). Refuted branches are rendered first-class, not collapsed.

6.5 The experiment-selection planner

Given a design, which experiment should one run next? The planner ranks each modelled observation o by expected information gain per unit cost, weighted by realizability:

$$\text{score}(o) = \mathbb{E}[\Delta|Z^*|] \cdot w_{\text{real}}/c_{\text{obs}}(o), \quad w_{\text{real}} = 1/\text{cost}(\text{unresolved edits}), \quad (2)$$

with speculative designs hard-excluded ($w_{\text{real}} = 0$). The two cost terms sit on different sides: observation cost c_{obs} (accurate mass 1, MS/MS 3, in committed relative units) in the denominator, and the super-additive realizability cost of Eq. (1) as the design-side weight. Because that weight is super-additive, resolving one of two stacked control edits drops the residual penalty non-linearly—so the planner reasons about the *joint resolution state* of a design, not merely the most-uncertain edge. The planner’s output is a decision tree over observation outcomes in which both children of every node are first-class: the refuted branch (the observation is inconsistent with the predicted product) carries as much information as the confirmed one.

The planner returns a typed triage verdict. VERIFIED-BY-TRIAGE: the modelled observables collapse $|Z^*|$ to near-unique. ELUCIDATION-REQUIRED: the modelled observables are exhausted and $|Z^*|$ is still large—the design must go to structure elucidation. OUT-OF-GRAMMAR: a mass observation refutes every legal program. *Naming the elucidation-required case is the contribution*: a triage tool that knows its own limits, and whose limits are typed and machine-checkable. Figure 2 contrasts two trees: 6-MSA collapses at accurate mass (a shallow tree—the planner doing its job), while the highly-reducing C₁₂ core BAB stays under-determined through mass and MS/MS (4050 $\rightarrow$ 25 $\rightarrow$ 23 candidates) and is routed to expression + NMR (the planner naming its limit).

This scoping is forced by the data, not chosen for modesty. All seven verified products in our corpus had their genome $\rightarrow$ structure link established by heterologous expression (with knockout or native expression) and NMR, with accurate mass and MS/MS serving as triage and detection—never as the sole structure-determining observable. The planner is therefore a metabologenomics *triage* tool: it answers “which observation collapses $|Z^*|$ given the genome,” not “how is this structure proven.” Evaluated retrospectively against this curated corpus it routes correctly—the small aromatics (orsellinic acid, 6-MSA, mellein, 6-hydroxymellein) verify by accurate mass, while the highly-reducing core is flagged elucidation-required—and on the design neighbourhood it steers attention to engineerable and frontier designs over speculative ones.

6.6 The architectural moat and the same-wall symmetry

Two claims carry the contribution. **First, it is architectural, not a capability we happen to have.** The control axis exists as a design dimension only for a sound executor that distinguishes per-cycle firing within a fixed domain set. “Reprogram the ketoreductase at cycle 3” is not a hard edit to a retrosynthesis tool (which works in chemical space), an enzyme-design tool (which works per protein), or a pathway tool (which ignores assembly-line grammar)—it is a sentence about an object none of them can represent. We are the missing layer between generative chemistry and

protein design, and the cost on the axis is the field’s measured difficulty (Eq. (1)). **Second, the same wall has opposite valence in the two directions.** The 100%→57% iteration-grammar collapse is, for *prediction*, a ceiling—where reconstruction fails (§3); for *design*, it is the frontier—the dimension along which the interesting molecules live. One measurement, a limitation read one way and the contribution read the other.

7 Limitations and outlook

[Stub — drafted after §6. *v0 NRPS/hybrid grammars; literature-curated edit tiers (the defensibility moat, curated from primary sources)*. Triage scoping, third placement: *the planner selects experiments and names elucidation-required clusters — it does not close the design–build–test loop*. Named *v1 extensions: observation-updated probabilistic realizability (tiers as success distributions, needing wet-lab build-outcome data); the per-cluster mining planner; and bacterial generalization (ClusterCAD manifold density) as the post-paper companion piece, where it lands as confirmation rather than competition.*]

8 Conclusion

[Stub.]

References

- R. J. Cox. Curiouser and curiouser: progress in understanding the programming of iterative highly-reducing polyketide synthases. *Nat. Prod. Rep.*, 40:9–27, 2023. doi:10.1039/D2NP00007E.
- K. M. Fisch, W. Bakeer, A. A. Yakasai, Z. Song, J. Pedrick, Z. Wasil, A. M. Bailey, C. M. Lazarus, T. J. Simpson, R. J. Cox. Rational domain swaps decipher programming in fungal highly-reducing polyketide synthases and resurrect an extinct metabolite. *J. Am. Chem. Soc.*, 133:16635–16641, 2011. doi:10.1021/ja206914q.
- M. M. Zdouc, K. Blin, et al. MIBiG 4.0: advancing biosynthetic gene cluster curation. *Nucleic Acids Res.*, 53:D678, 2025.
- Acyltransferases as tools for polyketide synthase engineering. *Antibiotics*, 7(3):62, 2018. PMC6164871. [authors to complete]
- Precursor-directed biosynthesis of erythromycin analogs by an engineered polyketide synthase. *Science*, 277:367, 1997. doi:10.1126/science.277.5324.367. [authors to complete]
- A polylinker approach to reductive loop swaps in modular polyketide synthases. PMID 18937219, 2008. [authors to complete]